

A Unified Framework for Human–AI Collaboration in Security Operations Centers with Trusted Autonomy

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This article presents a structured framework for Human-AI collaboration in Security Operations Centers (SOCs), integrating AI autonomy, trust calibration, and Human-in-the-Loop decision making. Existing frameworks in SOCs often focus narrowly on automation, lacking systematic structures to manage human oversight, trust calibration, and scalable autonomy with AI. Many assume static or binary autonomy settings, failing to account for the varied complexity, criticality, and risk across SOC tasks, considering human and AI collaboration. To address these limitations, we propose a novel autonomy tiered framework grounded in five levels of AI autonomy, from manual to fully autonomous, mapped to Human-in-the-Loop (HITL) roles and task-specific trust thresholds. This enables adaptive and explainable AI integration across core SOC functions, including monitoring, protection, threat detection, alert triage, and incident response. The proposed framework differentiates itself from previous research by creating formal connections between autonomy, trust, and HITL across various SOC levels, which allows for adaptive task distribution according to operational complexity and associated risks. The framework is exemplified through a simulated cyber range that features the cybersecurity AI Avatar, a RAG augmented LLM-based SOC assistant. The AI Avatar case study illustrates Human-AI collaboration for SOC tasks, reducing alert fatigue, enhancing response coordination, and strategically calibrating trust. This research systematically presents both the theoretical and practical aspects, as well as the feasibility of designing next-generation cognitive SOCs that leverage AI not to replace, but to enhance human decision-making.

CCS Concepts: • **Security and privacy** → **Security operations**; • **Computing methodologies** → **Artificial intelligence**; *Cognitive science*.

Additional Key Words and Phrases: Human–AI Collaboration, Security Operations Centers (SOCs), AI Autonomy and Trust, Cognitive SOC Framework, Cybersecurity Automation

1 Introduction

A Security Operations Center (SOC) serves as a centralized unit in cybersecurity, bringing together personnel, processes, and technology to protect an organization’s digital assets. As described in cybersecurity literature, an SOC functions as a proactive defensive hub, constantly monitoring, detecting, analyzing, and responding to cyber threats in real time within IT, operational technology (OT), and cloud settings. The SOC acts as the control center for cybersecurity tasks, integrating threat intelligence, security monitoring, incident response, and regulatory

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compliance to address emerging cyber risks. By utilizing advanced security analytics, automation, and machine learning, SOC's can detect anomalies, correlate security incidents, and mitigate threats before they develop into widespread breaches [22, 52]. Their importance is heightened in sectors at high risk, such as finance, healthcare, critical infrastructure, and government, where even a minor security incident can lead to dire operational and financial impacts. As cyber threats grow more advanced, organizations are increasingly dependent on SOC's to bolster their security measures, reduce risks, and ensure continuous business operations amid an evolving threat environment.

Amid the surge of state-sponsored cyberattacks and financially driven cybercriminal activities, SOC's have emerged as the primary defense against advanced and evolving threats. In 2024, cyberattacks worldwide increased by approximately 44%, with key targets being critical infrastructure organizations, particularly from state actors. Emerging cyber threats, such as cyber espionage and disruptive incursions into IT/OT, are now major strategic concerns [9]. The threat environment continues to evolve, with vulnerabilities rising by up to 17%. By 2027, cybercrime is projected to cause \$23 trillion in global damage [49], underscoring the essential need for strong cybersecurity operations. As adversaries adopt more sophisticated AI tools and techniques, organizations are increasingly committing resources to modernize SOC's, integrating automation, AI-powered analytics, and orchestration to boost efficiency. With global spending on cybersecurity expected to hit \$500 billion by 2030 [16], there is an intensified drive to strengthen SOC capabilities, especially through AI-based automation and analytics, to address these advancing cyber threats.

Despite the existing automation and AI-driven SOC capabilities, cybersecurity teams face overwhelming challenges in defending systems against cyber threats. A major issue is alert fatigue. SOC's generate thousands of security alerts daily, many of which are false positives. A global study found that SOC teams encounter 4,484 alerts per day, leading to 67% of alerts being ignored due to analysts being overloaded with many tasks. Studies show that up to 99% of SOC alerts can be false positives [10, 12, 20], contributing to analyst overload and 'alert fatigue'. Indeed, analysts often struggle with high false positive rates, cognitive fatigue, and a shortage of skilled personnel [5]. At the same time, modern attack campaigns can progress through multiple stages of the intrusion lifecycle, such as initial access, lateral movement, and data exfiltration, within minutes to hours, leaving limited time for manual investigation. Empirical studies of SOC operations show that analysts are frequently overwhelmed by large volumes of alerts and cannot investigate them in real time, resulting in delayed responses and missed threats [3, 56]. Consequently, the speed and scale of attack propagation can exceed the practical response capacity of human analysts alone, motivating the adoption of AI-driven automation for alert triage and incident investigation. In many modern attack campaigns, adversaries do not launch isolated attacks on separate systems; rather, they progress rapidly through multiple stages of the intrusion lifecycle within a single coordinated operation. For example, attackers may quickly move from initial access to lateral movement and ultimately to operational impact. Such rapid progression compresses the time available for analysts to investigate alerts and respond effectively.

SOC's rely on various security technologies, such as Security Information and Event Management (SIEM) systems, yet their efficiency is often limited by the need for extensive tuning and the handling of large volumes of uncorrelated alerts [10]. While SIEM platforms provide some automation, they still require significant human intervention, rendering cybersecurity operations largely manual and reactive [34, 45]. Security Orchestration, Automation, and Response (SOAR) platforms aim to streamline workflows, but their rule-based approaches struggle to adapt quickly to emerging attack patterns. Similarly, EDR and XDR solutions improve system visibility but continue to face challenges related to alert overload and rapidly evolving threats.

To address these challenges, organizations are increasingly integrating Artificial Intelligence (AI) and automation to enhance SOC operations. AI-driven security analytics can correlate data across multiple attack surfaces, automate incident response, and provide predictive threat intelligence. AI modeling techniques, i.e., Machine Learning (ML) and Deep Learning (DL), improve threat detection by identifying behavioral anomalies that deviate

from baseline activities, particularly for evolving threats. In practice, their effectiveness depends on contextual correlation and analyst validation to distinguish genuine threats from benign anomalies. Additionally, AI-powered automation enables SOC teams to prioritize, investigate, and contain threats more efficiently than traditional manual workflows [6, 10, 12]. A significant advancement in this domain is the application of Natural Language Processing (NLP) techniques, which enhance cybersecurity workflows by assisting in breach detection, threat identification, and scope analysis. LLMs have the potential to process unstructured security data, analyze logs, summarize incidents, and assist in real-time decision making, enabling natural language interactions that allow analysts to query security data and incident, related information and collaborate with AI-driven assistants for incident response. Despite these advancements, existing efforts lack a comprehensive framework for evaluating Human-AI collaboration across the full spectrum of SOC operations. Current approaches often overlook the balance between automation and human oversight, failing to integrate Human-in-the-Loop (HITL) strategies that ensure effective decision-making.

This work presents a Human-AI Collaboration Framework that focuses on augmentation, teaming, and adaptive automation for SOC operations. SOC functions are aligned with the NIST-based incident response lifecycle [43], highlighting the essential activities within a SOC. The framework formalizes a triadic model of autonomy, trust, and HITL collaboration, providing a conceptual foundation that links theoretical modeling with practical SOC operations. Effective Human-AI collaboration requires that autonomy levels dynamically align with the complexity of SOC tasks, the criticality of business operations, and the level of trust required for decision-making. The proposed five-level autonomy framework maps autonomy levels to SOC analyst roles, ranging from AI-augmented configurations with HITL to fully autonomous operations based on task complexity, thereby improving operational clarity and scalability.

Various Human-AI collaboration components of the framework are demonstrated through the SOC incident response lifecycle using an AI Avatar evaluated in a simulated cyber range that models real-world SOC dynamics. Developed under the ACDC project [16], the AI Avatar is a ChatGPT-based LLM augmented with retrieval-augmented generation (RAG) and security knowledge graphs, operating in a simulated SOC environment with SIEM, IDS, and firewalls. The system was developed using cyber range datasets generated over two years of wargaming scenarios, event logs, and Red/Blue Team exercises, enabling it to assist analysts in threat detection, triage, and incident response tasks. The results indicate improved SOC operational efficiency and enhanced trust based Human-AI decision making by supporting analysts in alert triage and incident investigation tasks.

The remainder of this paper is organized as follows: Section 2 reviews SOC operations, AI evolution, and challenges related to Human-AI collaboration. Section 3 introduces the foundational concepts of our framework. Section 4 presents the Human-AI Collaboration Framework, focusing on autonomy, trust, and task allocation. Related work and comparative analysis are provided in Section 5. Section 6 illustrates Human-AI collaboration through a case study. Section 7 discusses key contributions and limitations. Section 8 outlines future directions and concludes the paper.

2 Security Operations Centers: Functions, Challenges, and AI Evolution

SOCs Capabilities and Security Operations. Technological advancements and the growing sophistication of cyber threats have driven the continuous evolution of SOC. As adversaries increasingly employ advanced and evasive tactics, the need for rapid, accurate, and scalable threat detection and response has made the integration of AI and automation essential in modern SOC workflows.

Early SOC and Traditional SIEM Systems. Early SOC relied on Information and Event Management (SIEM) systems for aggregating and normalizing logs to detect threats using rule-based and signature-based methods [54]. These static systems struggled with high false positives and missed zero-day threats and Advanced Persistent Threats (APTs) due to a lack of adaptability [3, 35]. Statistical anomaly detection was also used but failed to

Table 1. Mapping of SOC functional domains to Tiered Analyst Roles and NIST CSF phases, highlighting systemic weaknesses and Human-AI limitations.

SOC Function (NIST CSF)	Key Challenges	Contributing Factors	Tier-Level Impact (1–4)
Identify – Asset Management	• Incomplete inventories • Poor system visibility • Hidden dependencies	• SIEM/EDR gaps • Log misconfiguration • Inconsistent aggregation	Tier 1–2: Log correlation gaps Tier 3–4: Limited situational overview
Protect – System Hardening	• Blind spots in defenses • Weak configuration hygiene • Limited adaptive controls	• Outdated rules • Insufficient training • Weak adaptability	Tier 1–2: Repetitive manual updates Tier 3–4: Policy misalignment with AI controls
Detect – Threat Monitoring	• High false positives • Fragmented telemetry • Low signal-to-noise ratio	• Alert overload • Analyst fatigue • Tool heterogeneity	Tier 1: Alert triage burden Tier 2–3: Escalation delays and context loss
Respond – Alert Handling	• Slow manual triage • Weak SOAR integration • Inefficient prioritisation	• Static prioritization • Partial knowledge bases • Weak automation	Tier 1–2: Slow validation Tier 3: Delayed containment Tier 4: Governance oversight gaps
Recover – Incident Remediation	• Slow recovery workflows • Patch/rollback delays • Compliance constraints	• Siloed workflows • Patch delays • Tier misalignment	Tier 1–2: Operational fatigue Tier 3–4: Strategic recovery bottlenecks

efficiently differentiate threats from benign anomalies, increasing alert fatigue [3]. Despite offering centralized visibility, SIEMs were hampered by log format inconsistencies and limited cross-platform correlation, leading to manual triage and investigation that delayed response and overburdened analysts.

Integrating SOAR, EDR, and XDR Technologies. To enhance operational agility and reduce manual workload, SOCs began integrating Security Orchestration, Automation, and Response (SOAR) platforms, Endpoint Detection and Response (EDR) tools, and more recently, Extended Detection and Response (XDR) solutions [17]. SOAR platforms introduced playbook-driven automation, executing predefined workflows in response to routine security events such as phishing, malware detection, or privilege misuse [24]. These playbooks use conditional logic to trigger contextual actions, streamlining response processes and reducing analyst intervention. EDR tools enhance endpoint visibility by monitoring activities, detecting abnormal behavior, and automating containment. XDR expands this by integrating telemetry across various environments for unified threat detection [17]. SOCs utilize API-driven orchestration to integrate SIEMs, firewalls, EDR, ticketing systems, and threat intelligence platforms, enabling automated actions like IP blocking, endpoint isolation, access revocation, and report generation with minimal input. Multi-source data correlation enhances incident enrichment, speeding investigations and reducing alert fatigue. Human oversight remains crucial for novel threats, emphasizing the need for AI systems with explainability and Human-in-the-Loop features to balance automation and accountability.

2.1 Cybersecurity Operations and Challenges in SOCs

The challenges discussed above arise from both operational and organizational limitations within SOC environments. To structure these issues, Table 1 summarizes the major SOC challenges and groups them according to their impact on key SOC functions, including monitoring, protection, threat detection, alert management, and incident response. The SOC analyst tier labels included in Table 1 are used only as an operational reference to illustrate how these challenges affect analysts with different operational responsibilities. Their formal definition and role within the proposed Human-AI Collaboration Framework are presented later in Section 3.4. This mapping provides the motivation for the tier-aware Human-AI collaboration model developed in the subsequent sections.

The following subsections examine these challenges in detail, beginning with core SOC functions and their associated operational limitations.

2.1.1 SOC Functions and Operational Challenges. We outline SOC operational hurdles and align them with the NIST Cybersecurity Framework (CSF) [43], a key industry standard for the incident response lifecycle. This mapping of SOC functional challenges to the incident response lifecycle offers practical insight into the issues SOC teams encounter, as elaborated below:

1. Asset Management. Effective asset identification is vital to SOC performance, yet SIEM and EDR tools have limitations in real-time asset discovery, especially in complex IT/OT environments. Manual processes and the transient nature of complex interdependent systems result in incomplete inventories, leaving endpoints unmanaged. The largest portion of effort and resources is dedicated to collecting and aggregating security data management. Inaccurate or missing logs, particularly from domain controllers, can obscure lateral movement by attackers, undermining detection efforts [54]. The absence of automated discovery and continuous asset validation increases vulnerabilities and impairs situational awareness, tying reliable log management to asset management. This is compounded by log capture errors and inadequate storage, posing security threats by expanding the attack surface.

2. Monitoring & Protection. Protection serves as a crucial function within a SOC, aiming to avert unauthorized system and data access. Its efficiency is frequently compromised by inflexible security configurations, inadequate upkeep, and insufficient employee training. Improperly set up firewalls or IDS may either overlook sophisticated threats or produce excessive false positives, resulting in security gaps [25]. Adversaries, such as APTs, exploit vulnerabilities through lateral movement or zero-day exploits. To counter this, SOC teams are using automation and machine learning. Tools like adaptive firewalls, anomaly-based intrusion prevention, and behavior analytics respond to new threats, reducing manual work and boosting data security. They can detect subtle activities, such as gradual data exfiltration, that traditional methods miss. However, enhanced protection requires advanced technology, skilled personnel, and AI support.

3. Threat Detection and Analysis. A typical SOC relies on SIEM platforms, EDR tools, and anomaly detection mechanisms to identify potential threats. However, these systems frequently generate large volumes of alerts, many of which are false positives, leading to analyst fatigue and reduced investigation efficiency [25, 54]. Automated detection systems may also struggle to recognize subtle attack patterns or correlate distributed indicators of compromise across multiple data sources. As a result, analysts must perform extensive manual investigations to validate alerts and determine their impact. These limitations, combined with the complexity of modern attack techniques such as lateral movement and stealthy malware, make effective threat detection and analysis a persistent operational challenge for SOC teams [22].

4. Alerts Management. SOC alert pipelines are inundated with alerts, with as much as 80% proving to be false positives owing to improperly configured detection rules and subpar models. This necessitates manual filtering by analysts, which postpones detection and response efforts, leading to fatigue and burnout [54, 59]. The absence of SOAR integration obstructs efficient prioritization and the correlation of data across various tools and resources. Alerts are meant to address cyber threats but are often plagued by tool inadequacies, human errors, and operational challenges, leading to fatigue and delays [3]. Alerts stem from security and endpoint tools and undergo analysis to differentiate threats from anomalies. Effective SOC management requires optimizing detection rules and automating responses with AI-driven prioritization and SOAR systems to reduce false positives. Enhancing SOC involves automation, agile intelligence, Human-AI collaboration, and real-time correlation to minimize noise and manual work.

5. Incident Response and Recovery. Effective incident response and recovery are critical for minimizing damage and restoring normal operations after a security breach. However, SOC teams face significant challenges that impede their incident response and recovery [13]. Organizations encounter difficulties in incident detection and

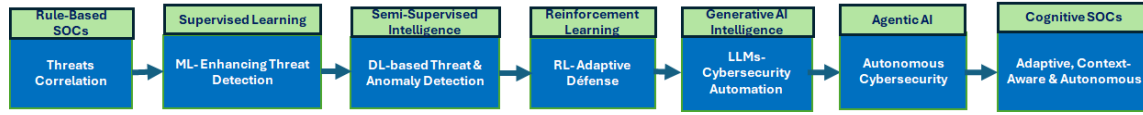


Fig. 1. AI Evolution for Cybersecurity in SOCs.

escalation due to inadequate monitoring, which prolongs exposure to vulnerabilities. According to the 2024 IBM Cost of a Data Breach Report, organizations took an average of 204 days to identify a breach and an additional 73 days to contain it, totaling 277 days duration that underscores the critical need for timely detection mechanisms [19]. Manual ticketing and isolated task management slow incident response due to inefficient workflows and poor coordination. Poor communication prolongs breach containment, escalating threats. High-severity incidents like ransomware require urgent responses, but outdated tools and organizational issues cause delays. Automated playbooks can reduce recovery times, yet concerns about AI reliability and legacy system compatibility hinder adoption. Legacy systems, reliant on outdated tech, complicate AI integration and pose security risks.

2.1.2 Human, Organizational and Technological Challenges. From a human perspective, SOCs face ongoing challenges of **operational overload, alert fatigue, and stress**. Analysts struggle with massive alert volumes, 90% being false positives [3, 19, 59], causing constant triage and neglect of true positives. Fatigue from irrelevant alerts increases the risk of missing real threats [19]. Stress from data overload and lack of **unified tools** further weakens SOC performance, burying critical signals and affecting morale, which leads to high turnover. With over 40% of SOCs **understaffed** [32], fewer analysts handle complex scenarios [54] and face skill gaps due to insufficient training and rapid threat evolution. Burnout and attrition are rising due to long hours, low autonomy, and a lack of recognition, resulting in higher errors and slower responses [3, 59]. SOC performance declines without support like escalation paths or flexible schedules. AI automation can improve detection and response times but faces challenges with explainability, adversarial threats, and legacy system integration. Trust in AI needs transparency and collaboration. Organizational barriers like outdated policies and silos slow responses during major incidents.

2.2 SOC Function Failures and Impacts

We analyze failures across key SOC functions, their underlying causes, and their impacts on cybersecurity operations. SOC effectiveness depends on the successful execution of core activities such as asset management, system protection, threat monitoring, alert handling, and incident response. However, these functions are frequently disrupted by operational limitations, including incomplete system visibility, delayed response processes, and difficulties in correlating large volumes of security data. Understanding these failure points is essential for improving SOC resilience and operational stability. Table 1 summarizes the major SOC functions and associated failure points, highlighting how operational gaps create vulnerabilities in security monitoring and response workflows. Each SOC functional category aligns with the phases of the NIST Cybersecurity Framework (Identify, Protect, Detect, Respond, Recover) [43], enabling a structured analysis of operational weaknesses. Operational failures such as incomplete asset identification and delayed response workflows weaken cybersecurity, generating high alert volumes that analysts must manually review. Many of these alerts are false positives, increasing workload, causing cognitive fatigue, and reducing threat detection and response efficiency [51]. Limited AI use weakens response capabilities [37], whereas AI-driven SOC systems can enhance threat detection, scalability, and resilience while reducing analyst workload [30, 42], as discussed below.

2.3 AI Evolution in Cybersecurity and SOCs

AI has evolved from rule-based systems in early SIEM and SOC platforms to advanced paradigms such as Machine Learning (ML), Deep Learning (DL), Generative AI, and, more recently, Agentic AI [45, 46]. Figure 1 illustrates

AI's expanding role in enhancing threat detection, classification, and response. As threats grow more advanced, organizations increasingly depend on SOC's to strengthen security measures, reduce risk, and ensure continuous business operations [22, 60]. These techniques are applied across the SOC lifecycle, including monitoring, detection, response, and recovery, each offering distinct advantages and operational challenges [45, 60].

Traditional SOC Rule-Based Systems. Before the emergence of data-driven AI techniques, SOC's primarily depended on rule-based systems for threat detection. These systems operated using predefined signatures, heuristics, and manually configured correlation rules to identify known attack patterns. While effective against well-understood threats, they lacked the flexibility to detect novel or zero-day attacks, as they could not learn or adapt from emerging threat data [34]. The reliance on manual rule creation also contributed to high false positive rates, increasing analyst workload and alert fatigue, which, in turn, delayed response times. These constraints highlight the inflexibility of static detection techniques, which have prompted the progression of AI in SOC's from machine learning to deep learning and the adaptive, AI-driven strategies discussed below.

Supervised Learning (ML): Enhancing Threat Detection. These are mostly ML architectures. Supervised learning models leverage labeled datasets to classify cyber threats, making them effective for tasks such as intrusion detection, malware classification, and spam filtering. Common techniques include Random Forests (RF), Support Vector Machines (SVM), Decision Trees (DT), and Deep Neural Networks (DNNs), which are foundational approaches in deep learning. For example, one study applied RF and Gradient Boosting Machines (GBM) in SOC-based intrusion detection, achieving high accuracy for known threats; however, performance degraded against novel attack patterns, underscoring the limitations of labeled-data dependence [22]. Similarly, hybrid models like CNN-LSTM, which combine convolutional feature extraction with sequence learning, have shown promise in malware detection but require frequent retraining to remain resilient against evolving threats [2]. Supervised models classify known threats efficiently in real-time but struggle with zero-day attacks, high labeling costs, model drift, and adversarial manipulation. They also face data imbalance and high false positives, especially concerning incoming threats.

Semi-Supervised Learning (DL): Threats & Anomaly Detection. Semi-supervised learning, often combined with deep learning architectures, is frequently employed in cybersecurity. This method utilizes a limited amount of labeled data together with a much larger set of unlabeled data, making it particularly beneficial in situations where labeled threat information is scarce. Prominent techniques include Graph Neural Networks (GCNs) and Variational Autoencoders (VAEs). A study introduced a GNN framework combining GCNs with a semi-supervised autoencoder [38], trained on both labeled and unlabeled data, improving threat detection recall by 18% over unsupervised methods [40, 60]. Semi-supervised architectures, such as VAEs with SVM classifiers, enhance anomaly detection but may flag rare, legitimate behaviors as outliers, increasing false positives. Thus, effective SOC's use correlation, contextual enrichment, or rule-based filtering to reduce false positives and prioritize alerts [37, 51]. These models are pretrained on unlabeled data through autoencoders, contrastive learning, and clustering, then fine-tuned with labeled data using standard loss functions. Semi-supervised learning enhances detection in low-label environments and identifies subtle anomalies signaling emerging security threats.

Reinforcement Learning (RL): Adaptive Defense Mechanisms. Reinforcement Learning models learn by interacting with the environment, which makes them valuable for developing adaptive cyber defense strategies. RL approaches such as Deep Q-Networks (DQN) models enable SOC agents to learn optimal responses through trial and error. Studies have shown that RL agents can outperform traditional rule-based intrusion detection systems, though they require extensive training and high-quality historical data [22, 41]. RL models typically employ Markov Decision Processes (MDPs) to maximize a security-based reward function and are trained using algorithms such as Policy Gradient Methods and Proximal Policy Optimization (PPO). In practice, RL supports:

- *Automated Incident Response:* Learning optimal responses to various attack scenarios to reduce response times.

- *Dynamic Resource Allocation*: Efficiently allocating defensive resources based on the current threat landscape.

Despite their promise, RL models face challenges in transparency (often functioning as “black boxes”), require substantial computational resources, and present integration challenges in real-world SOC environments.

Cybersecurity Automation with Generative AI. Generative AI, powered by advanced deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformers, is revolutionizing cybersecurity. By generating synthetic data, we enrich training datasets and improve the accuracy of threat detection models. Additionally, Generative AI supports secure software development and vulnerability discovery, reinforcing organizational resilience [21, 36]. Deep learning and generative architectures, including LSTM networks, model temporal dependencies in network traffic, while Transformer-based models further enhance detection accuracy in cloud and enterprise environments [4, 7, 28]. Models support knowledge graph creation through LLMs, enhancing threat intelligence and analysis. Transformer-based LLMs like BERT and GPT are fine-tuned for threat extraction and anomaly detection [29]. Practical uses include Google’s Sec-PaLM [11], using Mandiant and VirusTotal data, and Microsoft’s Security Copilot [33], integrating with SIEM and XDR for better investigation, threat hunting, and automated SOC responses.

Agentic AI. Agentic AI, grounded in Large Language Models (LLMs), Chain-of-Thought (CoT) reasoning, and coordinated Agentic Workflows, enables a shift from reactive to proactive cybersecurity operations in future SOCs [21, 27, 36]. These systems autonomously analyze large-scale telemetry and threat intelligence data to generate real-time, scalable insights. CoT enhances transparency in decision-making and enables agents to perform complex tasks, such as identifying IOCs and assessing incident impact [23]. Agentic Workflows orchestrate task-specific agents across the incident response lifecycle, from detection to recovery, while LLM-driven agents proactively forecast vulnerabilities and recommend actionable mitigation strategies.

Towards Cognitive Security Operation Centers. The evolution of AI is fundamentally transforming cybersecurity, shifting from static, rule-based tools to adaptive, context-aware systems that actively support threat detection, anomaly analysis, and incident response. Modern SOCs increasingly employ natural language processing (NLP), machine learning/deep learning (ML/DL), and generative AI across large language models (LLMs) and multi-agent frameworks. These advancements have given rise to *Cognitive SOCs*, where AI assists analysts by correlating threat data, triaging incidents, and automating repetitive tasks. Such systems enhance detection accuracy, reduce response time, and increase operational resilience. Early SOC automation was reactive, requiring manual rule adjustments and lacking adaptability for zero-day exploits, APTs, and AI-driven attacks. This highlights the need for structured Human–AI collaboration to improve scalability, interpretability, and expertise, balancing autonomy with oversight for proactive, resilient cybersecurity. Building on the evolution of AI capabilities in SOCs environments and in response to both functional and organizational challenges, the following section introduces foundational concepts essential for effective Human–AI collaboration, which are central to the design and operation of Cognitive SOCs.

As a practical implementation of AI-assisted SOC operations, we use *CyberAlly*, an AI-enabled SOC system developed within the ACDC cyber range [23]. It leverages a GPT-class LLM with retrieval and security knowledge graphs for alert triage, threat analysis, and incident response. Evaluation is based on simulated SOC wargame datasets (see Section 6 and Appendix A).

3 Theoretical Modeling of Human–AI Collaboration in SOCs

Human–AI collaboration presents unique challenges in high-stakes domains such as cybersecurity, where decision-making requires both rapid response and contextual judgment in fast-paced and uncertain environments. While AI offers scalability and speed [10, 50], it often lacks explainability and situational awareness, key traits of human cognition. Achieving synergy between AI autonomy and human oversight is difficult due to conflicting requirements: trust must be earned through consistent performance, while excessive automation risks alienating human

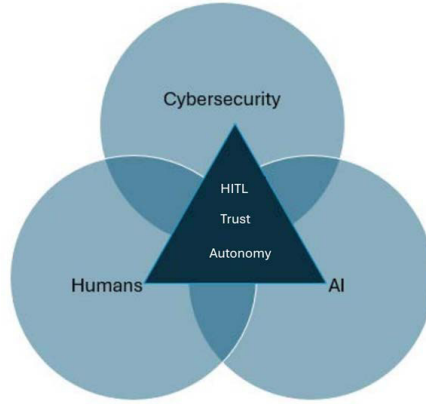


Fig. 2. Human-AI Collaboration in SOCs: The triadic relationship between Autonomy (A), Human-in-the-Loop (HITL), and Trust (T).

analysts. This tension complicates harmonizing autonomy, trust, and collaboration. Human-AI collaboration in SOCs relies on three core pillars that combine complementary strengths: Autonomy, Human-in-the-Loop (HITL), and Trust (Figure 2).

3.1 Autonomy

AI autonomy refers to a system's ability to independently execute tasks and make decisions without continuous human intervention. Trust (T) represents the analyst's confidence in AI recommendations and is treated as a normalized variable in the range $[0,1]$; its formal definition is provided in Section 3.3. Formally, autonomy A is modeled as a function of task complexity (C), operational risk (R), and calibrated trust (T), where C and R exert constraining pressure on autonomy, and T provides enabling support. Autonomy in SOC operations spans a spectrum from fully human-controlled (low autonomy) to fully independent execution (high autonomy). This spectrum also reflects practical differences in system responsiveness: while AI-assisted detection and response in optimized ML systems can react in near real-time under operational constraints [61], Large-scale SOC automation using LLMs and RAG pipelines typically responds within seconds to minutes, depending on investigation depth. Balancing these autonomy levels is essential to maintain efficiency while mitigating risks such as automation complacency and reduced contextual awareness [1, 53]. We define autonomy as a scalar $A \in [0, 1]$, modeled as:

$$A = 1 - (\lambda_1 C + \lambda_2 R)(1 - T) \quad (1)$$

The weights λ_1 and λ_2 control the influence of task complexity (C) and operational risk (R) on the autonomy decision. Larger λ_1 increases AI assistance for complex investigations, whereas larger λ_2 prioritizes human oversight for high-risk incidents. Balanced weighting ($\lambda_1 = 0.5$, $\lambda_2 = 0.5$) was used to reflect typical SOC decision practices and produced stable autonomy behavior in CyberAlly evaluations.

In an SOC, the model variables are derived from operational data and normalized to the range $[0, 1]$ to ensure comparability across metrics. Investigation complexity (C) is estimated from correlated alerts, investigation depth, and analyst effort. Operational risk (R) is derived from incident severity and threat likelihood based on SOC alert classification. Trust (T) is computed using the trust model defined in Eq. 3. The weights λ_1 and λ_2 in Equation 1 ($\lambda_1 + \lambda_2 = 1$) represent the relative influence of task complexity and operational risk on the autonomy decision. These weights were derived from expert judgments and validated through sensitivity analysis (Section 6), mirroring SOC decision patterns observed during evaluations with CyberAlly, an AI-assisted SOC platform

developed within the ACDC cyber range and described in Section 6 [23], used to assess SOC decision-making in simulated wargame scenarios.

3.2 Human-in-the-Loop

The Human-in-the-Loop (HITL) paradigm integrates human judgment into AI-driven workflows, which is vital during uncertain, novel, or high-stakes incidents [6, 30]. As autonomy increases, human involvement tends to decrease [37], forming a critical inverse relationship for balanced and accountable decision-making. We define HITL involvement $H \in [0, 1]$ as the effective proportion of decision effort requiring analyst verification or intervention. In the proposed framework, H is not an independent variable; rather, it is defined as the complement of autonomy, where $H = 1$ denotes full human control and $H = 0$ denotes full AI autonomy. Accordingly,

$$A = 1 - H \quad (2)$$

Combining Eq. (1) and Eq. (2), the effective human involvement induced by the model is

$$H = (\lambda_1 C + \lambda_2 R)(1 - T).$$

Autonomy (A) increases as human involvement decreases; $A = 1 - H$ reflects this. Effective HITL designs help prevent automation bias, maintain situational awareness, and ensure ethical oversight [30, 37].

3.3 Trust

Trust reflects a security analyst's confidence in AI transparency, competence, and reliability. Properly calibrated trust prevents both overtrust, which may lead to neglect, and undertrust, which limits effective AI utilization [1, 47, 58]. To operationalize trust, its components are derived from observable SOC interaction data and normalized to the range $[0, 1]$. Explainability (E) is approximated using structural properties of AI-generated outputs (e.g., relative explanation length), capturing the level of interpretability provided to analysts. Performance (P) is derived from analyst feedback scores reflecting the perceived reliability of AI recommendations. Uncertainty (U) is computed from the variability of analyst confidence ratings, representing inconsistency in AI outputs. For example, during a ransomware investigation, the firewall reports suspicious outbound connections, but the corresponding endpoint telemetry is unavailable, resulting in incomplete and inconsistent SOC logs. The conflicting security evidence increases uncertainty (U), reducing trust in the AI's recommendation to isolate the affected host. These measures ground trust in empirical evidence of interaction rather than in assumed properties. Trust (T) is modeled as a function of explainability, performance, and uncertainty:

$$T = \alpha_1 E + \alpha_2 P + \alpha_3 (1 - U) \quad (3)$$

where E , P , and U are defined and derived as detailed in Appendix A. The parameters α_1 , α_2 , and α_3 represent the relative influence of each component, with $\alpha_1 + \alpha_2 + \alpha_3 = 1$. Higher explainability and consistent performance increase trust, while greater uncertainty reduces it. Trust acts as an enabling factor for autonomy, supporting greater delegation of decision-making to AI systems as analyst confidence increases.

3.4 Human SOC Role Tiers

The responsibilities of SOC analysts can be systematically aligned with the functional domains defined in the NIST Cybersecurity Framework (CSF) 2.0 [43]. Building on the operational challenges discussed in Section 2.1, this section formalizes the SOC analyst tiers that were introduced as an operational reference in Table 1. These tiers establish the basis for integrating AI autonomy and Human-in-the-Loop (HITL) decision-making across SOC responsibilities. Tiered structuring improves operational efficiency, clarifies the interaction between human analysts and AI-driven systems, and supports a structured, proactive approach to cybersecurity operations [39, 55].

- *Tier 1* primarily supports the *Identify* and *Protect* functions. Analysts at this level are responsible for system inventory, vulnerability identification, baseline configuration, and the enforcement of preventive controls. They perform real-time monitoring, initial alert triage, and escalate incidents as needed.
- *Tier 2* aligns with the *Detect* and *Respond* functions, involving detailed threat investigation, correlation of security events, and the execution of containment strategies. Tier 2 analysts often act as incident handlers, coordinating with other teams to manage security breaches.
- *Tier 3* spans the *Detect* and *Recover* domains. These senior analysts optimize and refine detection strategies and rules, lead threat hunting activities, conduct malware and forensics investigations, and assist in the development of recovery procedures.
- *Tier 4* corresponds to the *Govern* and *Recover* functions. Responsibilities include policy oversight, risk management, compliance enforcement, and high-level decision-making during complex incidents such as ransomware outbreaks or critical infrastructure attacks. This tier is often engaged in strategic planning and aligning SOC goals with enterprise cybersecurity governance

We establish SOC analyst roles that clarify responsibilities across operational tiers and provide a foundation for integrating AI systems into SOC workflows. This structure supports tier-aware AI autonomy models that enable Human-in-the-Loop (HITL) decision-making, achieving effective automation while preserving essential human oversight in critical operations.

Hybrid Tier Adaptation. In resource constrained SOC¹s Strict tier separation is often impractical. The framework, therefore, supports hybrid roles that merge Tier 1–2 responsibilities under a unified analyst profile. Autonomy and HITL scaling adjust dynamically to the analyst’s workload and trust calibration rather than a fixed hierarchy, ensuring adaptability across varying staffing conditions.

The following section presents a Human-AI Collaboration Framework for SOC¹s, building on the concepts outlined above to improve decision-making, situational awareness, and operational resilience.

4 Human-AI Collaboration Framework for Security Operations Centers

Building upon the theoretical model established in Section 3, this section operationalizes that foundation into a practical Human-AI Collaboration Framework for SOC¹s. The framework translates autonomy, trust, and human involvement into a structured system to address alert fatigue, false positives, and incomplete incident responses. It offers a multi-layered strategy for integrating AI with human analysts using defined autonomy levels, feedback, and trust pathways, fostering reproducible, adaptive collaboration in modern SOC¹s.

SOC¹s face alert fatigue, false positives, and human burnout, often lacking a complete incident response cycle for effective threat management and decision-making. Effective AI integration, therefore, requires a structured strategy, which this framework provides through a multi-layered approach to human-AI collaboration in SOC environments. Figure 3 illustrates the proposed framework. The bottom layer defines five autonomy levels (Levels 0–4), aligned with HITL, Human-on-the-Loop (HOTL), and Human-out-of-the-Loop (HOO TL) roles. As trust in AI systems grows, autonomy can be scaled incrementally, ensuring that governance, accountability, and operational safety remain intact. This layered structure supports flexible, modular deployment across diverse SOC environments. From the top layer, the framework organizes SOC functions across four tiers: System Security, Alerts Management, Threat Management, and Response & Recovery, mapped to increasing levels of AI autonomy and corresponding human oversight roles. Routine, high-volume tasks are increasingly delegated to AI agents (Levels 1–3), while analysts retain strategic control over critical decisions. At the core of the framework is a symbiotic interaction between human and AI capabilities. AI contributes to large-scale data processing, predictive analytics, and autonomous decision-making, while humans provide contextual reasoning, business awareness,

¹SOCs, such as Managed Security Service Providers (MSSPs), which operate with limited staffing, automation, and budget.

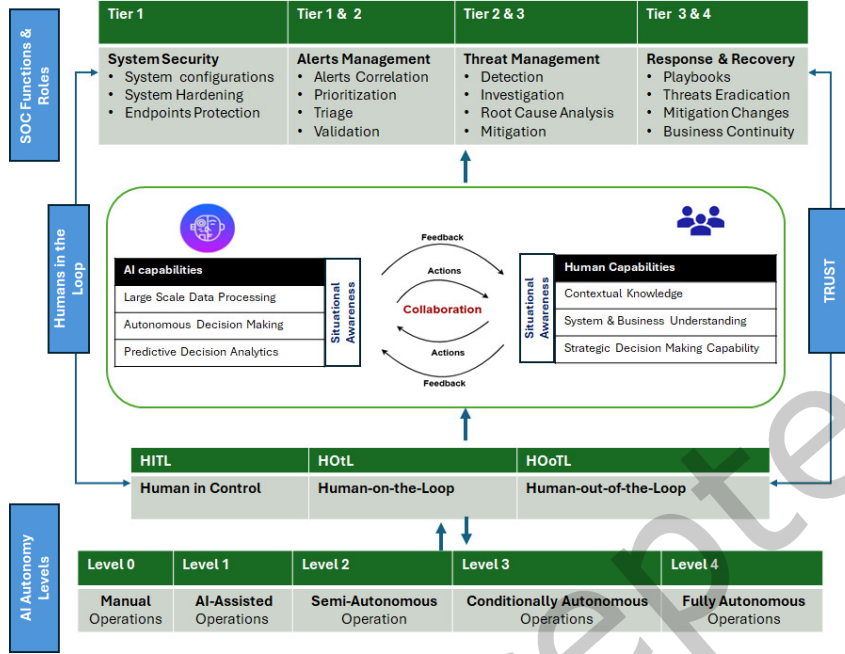


Fig. 3. The Human-AI Collaboration Framework in SOC functional tiers to AI autonomy levels (0–4) and human roles from HITL to HOoTL, supporting decision-making, learning, and automation in cybersecurity through situational awareness and trust.

and ethical judgment. Feedback loops and transparency mechanisms (e.g., explainability panels, uncertainty scoring) support trust calibration and autonomy scaling.

4.1 Autonomy Calibration

The autonomy level (A^*) for a given SOC task is determined by combining task complexity (C), operational risk (R), and analyst trust in AI recommendations (T). To incorporate trust into the autonomy decision, we define a trust-modulated autonomy function:

$$A^* = f(C, R, T) = 1 - (\lambda_1 C + \lambda_2 R)(1 - T), \quad (4)$$

where λ_1 and λ_2 are analyst-configurable weights representing the relative importance of task complexity and operational risk. While Eq. (1) defines the general autonomy formulation, A^* represents its calibrated instantiation under specific parameter settings and operational conditions in the CyberAlly case study. The variables C , R , and T follow the formal definitions introduced in Section 3.3, with their empirical computation and normalization detailed in Section 6.2 and summarized in Table 7. To enable dynamic autonomy adaptation during operations, a feedback controller updates the autonomy level as:

$$A_{t+1} = A_t + \beta(\Delta T - \Delta R), \quad (5)$$

where β is a sensitivity coefficient controlling how changes in trust and operational risk influence autonomy. Here, $\Delta T = T_{t+1} - T_t$ represents changes in analyst trust, while $\Delta R = R_{t+1} - R_t$ captures changes in incident risk. Increasing trust and decreasing risk lead to higher autonomy, whereas declining trust or increasing risk results in

reduced autonomy and greater human oversight. For instance, during a routine phishing investigation, increasing analyst trust (ΔT) and decreasing operational risk (ΔR) allow routine decisions to be handled at higher autonomy levels while preserving human oversight for higher-risk tasks. Table 2 summarizes the autonomy levels and their mapping to representative SOC decision workflows.

Table 2. AI Autonomy Levels in SOC and Corresponding Ranges. The autonomy range (A) reflects the degree of AI independence in SOC decision-making, aligned with the Human-AI Collaboration Framework.

Level 0 Autonomy	Level 1 Autonomy	Level 2 Autonomy	Level 3 Autonomy	Level 4 Autonomy
$A \approx 0$	$A \approx 0.2-0.4$	$A \approx 0.4-0.6$	$A \approx 0.7-0.8$	$A \approx 0.9-1.0$

Level 0 – Manual Operations (No Autonomy). Analysts manually handle all security alerts and responses without automation. *Example: Analysts manually review raw logs in SIEM systems, manually correlate firewall events, and execute incident response tasks (such as blocking IPs or isolating endpoints) individually, causing delays and increasing the potential for human error.*

Level 1 – AI-Assisted Operations (Decision Support). AI assists by providing alert prioritization or recommending investigative actions through SIEM dashboards or ML models. Analysts maintain decision control. *Example: A SIEM employs machine learning to score alerts by severity. Analysts manually review high-severity alerts and approve recommended actions suggested by the system, such as manually resetting passwords or quarantining suspicious files.*

Level 2 – Semi-Autonomous Operations (AI Acts with Human Approval). AI systems, such as SOAR tools, automate routine detections and propose response workflows, awaiting explicit analyst approval for critical actions. *Example: A SOAR platform automatically triggers endpoint isolation workflows upon detecting malware, but it awaits analyst confirmation before isolating critical servers or privileged accounts. Analysts receive concise notifications and quickly approve or reject the recommended actions.*

Level 3 – Conditionally Autonomous Operations (Human-in-the-Loop). Advanced AI agents [27] autonomously execute defined security tasks, involving analysts only in exceptional scenarios. Analysts monitor these systems in a supervisory manner (human-on-the-loop). *Example: AI-driven SOAR and security analytics tools autonomously detect phishing emails, remove them from mailboxes, block malicious IP addresses, and isolate compromised endpoints. Analysts supervise these autonomous processes through periodic reviews or intervene only when AI flags uncertainties or novel attack patterns.*

Level 4 – Fully Autonomous Operations (Minimal Human Oversight). Highly autonomous AI agents manage most SOC functions with minimal human involvement, aligning with emerging *Agentic AI* paradigms that enable end-to-end orchestration and adaptive decision-making in cybersecurity operations [21, 27]. Analysts primarily focus on governance, auditing, and policy oversight to maintain accountability and transparency [1]. *Example: An AI-driven SOC platform automatically ingests threat intelligence, conducts proactive threat hunting, autonomously contains ransomware by isolating endpoints, initiates recovery actions, and generates incident reports while continuously updating detection models. Analysts periodically review AI decisions and provide strategic oversight without intervening in routine operations [22].*

The following subsection introduces Human-in-the-Loop (HIL) integration to balance autonomy, trust, and human oversight.

4.2 SOC AI: Human-in-the-Loop

The HITL paradigm embeds human judgment into AI-driven workflows, which is especially critical during ambiguous, novel, or high-risk cyber incidents [6]. Building upon the theoretical definition introduced in Section 3 (Eq. 3) and the autonomy calibration process in Section 4.1 (Eqs. 4–5), we extend the static HITL formulation to

its dynamic operational form. We define the *effective HITL involvement* H_t at time t as inversely related to the dynamically adjusted autonomy level A_t :

$$H_t = 1 - A_t = 1 - [1 - (\lambda_1 C_t + \lambda_2 R_t)(1 - T_t)] \quad (6)$$

where C_t , R_t , and T_t denote task complexity, risk, and trust at time t , respectively. Hence, as trust T_t increases or operational risk R_t decreases, autonomy A_t rises, and the corresponding human involvement H_t proportionally declines, maintaining accountability while enabling rapid response.

Levels of Human–AI Involvement in SOC. We define three levels of HITL to capture varying degrees of human involvement across the SOC autonomy spectrum:

- (1) *Human-in-Control (full HITL)*. Humans make all operational decisions; AI provides support (e.g., scoring alerts or suggesting actions). Common at *Autonomy Level 1*, as seen in SIEM-driven SOC where analysts manually investigate and respond.
- (2) *Human-on-the-Loop (HOtL)*. AI executes predefined tasks under conditional human oversight. Analysts monitor operations and intervene when alerts require contextual judgment. Typical of *Autonomy Levels 2–3*, such as SOAR playbooks that automate response with analyst escalation for complex cases.
- (3) *Human-out-of-the-Loop (HOOtL)*. AI autonomously manages routine tasks with minimal human interaction, primarily for audit or policy verification. Found in *Autonomy Level 4* settings where AI handles end-to-end threat response (e.g., malware isolation, access revocation).

Relying too much on full HITL slows operations and hinders AI efficiency *automation bottlenecks*. On the other hand, full HOOtL settings may result in AI actions without human awareness, leading to unnoticed misclassifications or new threats *loss of oversight*. Effective HITL needs *trust and training*; analysts should understand AI, oversee decisions, and intervene when needed [30, 37]. *Tier-1 analysts* work in HITL or HOtL modes, handling alerts and initial responses with AI aid. *Tier-2 analysts* are Human-on-the-Loop, managing workflows and AI decisions. *Tier-3 analysts*, at the top, operate in a Human-out-of-the-Loop role, focusing on threat hunting, AI oversight, and ensuring ethics and policy adherence. To establish a reliable Human-AI collaboration within SOC, dynamic HITL levels are tailored to task complexity, risk, and AI trust. This is integrated with autonomy calibration (see Section 4.1) and trust management (refer to Section 3.3).

Autonomy Switching Triggers. Building on the calibration functions in Eqs. 4–5, autonomy transitions are dynamically governed by trust (T) and uncertainty (U). If $U > 0.4$ or $T < 0.5$, the system reverts to HITL (Levels 1–2); when $T > 0.8$ and $U < 0.2$, autonomy escalates to Levels 3–4. The corresponding decision thresholds and examples of their application across SOC roles are summarized in Table 3.

Table 3. Autonomy switching triggers mapped to trust (T), uncertainty (U), and corresponding SOC roles.

Condition	Adjustment	SOC Role Example
$T < 0.5$ or $U > 0.4$	Downgrade autonomy	Tier-1 analyst: manual triage and verification
$0.5 \leq T \leq 0.8, U \leq 0.3$	Maintain conditional autonomy	Tier-2 co-monitoring: shared control between AI and analyst
$T > 0.8$ and $U < 0.2$	Escalate autonomy	Tier-3–4 oversight: AI-led operations with human supervision

4.3 Trust: Confidence Pillar Between Humans and AI

Trust represents the third foundational pillar of the Human-AI Collaboration Framework, mediating the relationship between autonomy and Human-in-the-Loop involvement. As autonomy increases, direct human oversight diminishes, making calibrated trust essential for safely delegating operational decisions to AI systems in SOC. As established in the autonomy switching logic (Table 3), trust T functions as the principal regulatory variable governing transitions between operational levels of autonomy. When trust and confidence exceed defined thresholds, autonomy scales (Levels 3–4) upward; when trust declines or uncertainty increases, the system reverts to Human-in-the-Loop supervision (Levels 1–2). This dependency highlights trust calibration as a key mechanism for maintaining balanced and adaptive Human-AI collaboration.

Dynamic Trust Update Function. Trust evolves continuously as the AI demonstrates reliability and transparency. We define a time-dependent update rule:

$$T_t = T_{t-1} + \Delta T(E, P, U) \quad (7)$$

Here $\Delta T = \eta_1 \Delta E + \eta_2 \Delta P - \eta_3 \Delta U$, with $\Delta E = E_t - E_{t-1}$, $\Delta P = P_t - P_{t-1}$, and $\Delta U = U_t - U_{t-1}$. The weighting parameters η_1 , η_2 , and η_3 determine the relative influence of explainability, performance history, and uncertainty on trust evolution, respectively. Specifically, η_1 weights the contribution of explainability, η_2 weights the contribution of performance history, and η_3 weights the reduction in trust associated with increasing uncertainty. For example, during consecutive AI-assisted investigations, positive changes in explainability ($\Delta E > 0$) and performance ($\Delta P > 0$), together with a reduction in uncertainty ($\Delta U < 0$), increase trust ($\Delta T > 0$). Conversely, declining explainability or performance, or increasing uncertainty, reduces trust. This continuous trust update enables real-time trust recalibration and directly informs the autonomy controller in Eq. 5.

At Levels 0–1 autonomy, trust has limited influence as analysts retain control and validate AI outputs, with AI mainly supporting tasks such as alert prioritization. As trust increases and task complexity and risk remain manageable, systems can operate at higher autonomy levels (Eq. 2). At Levels 2–3, calibrated trust is critical, as overtrust can lead to automation bias, whereas undertrust reduces efficiency [1, 58]. Balancing trust requires *Explainable AI (XAI)* to make AI reasoning interpretable, particularly in high-alert environments [1, 51]. However, interpretability limits and adversarial vulnerabilities in black-box models remain key challenges [1, 53]. Trust, together with autonomy and HITL oversight, forms the foundation for secure and adaptive Human-AI collaboration in SOC.

Trust-Conflict Reconciliation. When AI explanations diverge from analyst intuition, the system triggers an *explanation-reconciliation mode*. Analysts can challenge or annotate AI outputs, prompting refined justifications with added context and uncertainty. This process supports calibrated trust and aligns AI reasoning with expert judgment.

4.4 Unified Representation of Autonomy, HITL, and Trust

The core elements essential for successful Human-AI collaboration within SOC are *Autonomy (A)*, *Human-in-the-Loop (H)*, and *Trust (T)*, and they are closely interrelated. Table 4 offers a comprehensive operational framework that demonstrates how varying levels of HITL align with SOC tasks, task complexity, AI autonomy, and their practical effects. This framework underscores the need for a dynamic equilibrium between human oversight and machine autonomy, emphasizing how trust and task complexity influence the level of human engagement in AI-supported SOC activities. The interaction of these elements dictates the deployment of AI systems, the control retained by human analysts, and the perceived reliability of AI. Autonomy specifies how independently AI operates; HITL indicates the level of human oversight; and trust influences both, facilitating the transition between human-led and AI-driven decision-making.

Table 4. Operational mapping of Human-in-the-Loop (HITL) levels with task complexity and autonomy in SOC operations.

HITL Level	SOC Task Example	Task Complexity	AI Autonomy	Operational Implications
Full HITL	Investigating zero-day threats	High	Low	Manual control
Partial HITL	Quarantining suspicious malware	Medium	Medium	Shared oversight
HOtL	Auto-classifying phishing emails	Low-Medium	High	AI-led with review
HOoTL	Blocking malicious IPs; patching	Low	High	Full automation

To operationalize this triadic relationship, Table 5 maps representative SOC tasks to autonomy (A), human involvement ($H = 1 - A$), and required trust (T). High-autonomy tasks (e.g., blocking malicious IPs) require minimal human input but high trust, while novel tasks such as zero-day investigations demand low autonomy and greater oversight. Intermediate tasks, such as phishing classification, benefit from shared control. Aligning these factors with task characteristics improves SOC efficiency while preserving accountability and human judgment.

Table 5. Triadic mapping of autonomy (A), HITL (H), and trust (T) across SOC tasks.

Legend: A = Autonomy (0 = manual, 1 = full autonomy); H = Human-in-the-Loop (inverse of A); T = Required trust level for safe AI delegation.

SOC Task	Autonomy (A)	HITL (H)	Trust (T)
Blocking a known malicious IP	High (0.8–1.0)	Low (0–0.2)	High
Phishing email classification with human review	Moderate (0.4–0.7)	Moderate (0.3–0.6)	Medium
Investigating zero-day exploit	Low (0.1–0.3)	High (0.7–0.9)	Low

4.5 Situational Awareness: Enabling Shared Human–AI Context

Situational Awareness (SA) is a foundational component of the proposed Human–AI Collaboration Framework for SOC, enabling shared, context-sensitive decision-making across varying levels of AI autonomy and analyst roles. Within HITL configurations, SA ensures that AI-generated insights align with human mental models, preserving accountability and interpretability in operational decisions. As Aref et al. [6] emphasizes, shared awareness between humans and AI enhances team cognition and improves detection accuracy, particularly for stealthy or ambiguous threats. Following Endsley’s three-stage model of perception, comprehension, and projection [15], SA in SOC depends on collaborative threat detection, interpretation, and predictive analysis between humans and AI agents. Aref et al. [6] further stress that AI should not operate in isolation but should communicate its reasoning and uncertainty clearly to assist analyst decision-making. Key challenges include automation bias, which causes analysts to over-rely on AI and suppress critical judgment [51], and cognitive overload from excessive alerts that reduce vigilance [37]. Both conditions degrade situational awareness and hinder the identification of high-priority threats. To mitigate this, AI systems must adapt information delivery according to cognitive load, contextual relevance, and urgency [6].

Adaptive Update Throttling. To prevent cognitive overload from excessive AI notifications, the framework implements adaptive relevance scoring and dynamic rate-limiting. Low-confidence or redundant alerts are aggregated into periodic summaries, while high-confidence or time-critical insights are surfaced immediately.

This mechanism preserves situational clarity and ensures analysts remain focused on the most relevant operational context.

Three Dimensions of Collaborative Awareness. Building upon the A²C framework by Baruwat Chhetri et al. [10], SA is conceptualized as an interactive feedback loop between AI capabilities, such as autonomous analytics and large-scale data processing, and human expertise, including domain knowledge and contextual understanding. This model defines three interconnected dimensions of SA: *self-awareness*, *teammate awareness*, and *world awareness*.

1. **Self-awareness.** Both humans and AI agents must recognize their own limitations and confidence boundaries. Humans should acknowledge factors like fatigue or skill gaps, while AI must assess its uncertainty and defer to human supervision when confidence thresholds are low.
2. **Teammate awareness.** This involves mutual understanding between human analysts and AI systems, ensuring awareness of each other's roles, cognitive states, and workloads. Such awareness fosters trust, coordination, and transparency across collaborative SOC workflows.
3. **World awareness.** Both agents maintain a coherent, up-to-date understanding of the operational environment, including threat landscapes, system status, organizational priorities, and ongoing response measures, enabling adaptive and context-aware decision-making in real time.

In our framework (see Figure 3), situational awareness emerges as a mediator of collaboration, dynamically shaped through continuous feedback between human and AI actions. It enables a seamless transition across HIL configurations, aligned with varying levels of autonomy and SOC tiers. This alignment facilitates role-adaptive information exchange, enhances analyst trust, and supports timely decision-making in high-pressure environments.

4.6 SOC Role Tiers: Functional Responsibilities and AI Alignment

This section outlines SOC roles aligned with AI within our framework. Security Operations Centers use a multi-tiered approach to assign analyst duties based on complexity, expertise, and risk [54]. Tiers 1 through 4 align with the NIST CSF functions [43], forming the base of our Human-AI Collaboration Framework. This setup integrates AI autonomy with human oversight [1] for effective cyber defense. In a ransomware attack starting with an endpoint breach, SOC responses are managed across these tiers, using varying levels [22] of AI and human collaboration. While ransomware is used here for illustration, the framework also generalises to other incident categories such as DDoS and APT-style lateral-movement attacks, which follow the same tier-aligned Human-AI collaboration principles.

Tier 1: System Hardening and Alert Monitoring (Identify/Protect). Attacks are initially detected by Tier-1 analysts monitoring security telemetry and endpoint behavior [25, 30]. At low AI autonomy (Levels 0–1), analysts manually inspect alerts using SIEM and EDR systems. As autonomy increases (Level 2), AI tools assist in event correlation, anomaly detection, and alert prioritization. In HITL mode, analysts validate pre-processed alerts and initiate playbooks, ensuring quick triage without compromising judgment.

Tier 2: Threat Investigation and Coordinated Mitigation (Detect/Respond). Upon escalation, Tier-2 analysts investigate the threat and initiate containment procedures, such as isolating compromised systems [19, 51]. At autonomy Levels 2–3, AI enriches investigations with contextual data and suggested mitigation steps. Analysts supervise SOAR workflows in HOTL mode, adapting automated actions to the operational context.

Tier 3: Proactive Threat Hunting and System Optimization (Detect/Recover). Tier-3 analysts explore the attack's scope, identifying the attacker's lateral movement and persistent footholds [54]. AI systems with higher autonomy (Levels 3–4) proactively detect similar anomalous behavior across the network and generate threat hypotheses [27]. Analysts validate these insights and optimize and refine detection strategies and correlation rules, improving detection effectiveness through continuous analyst-AI feedback.

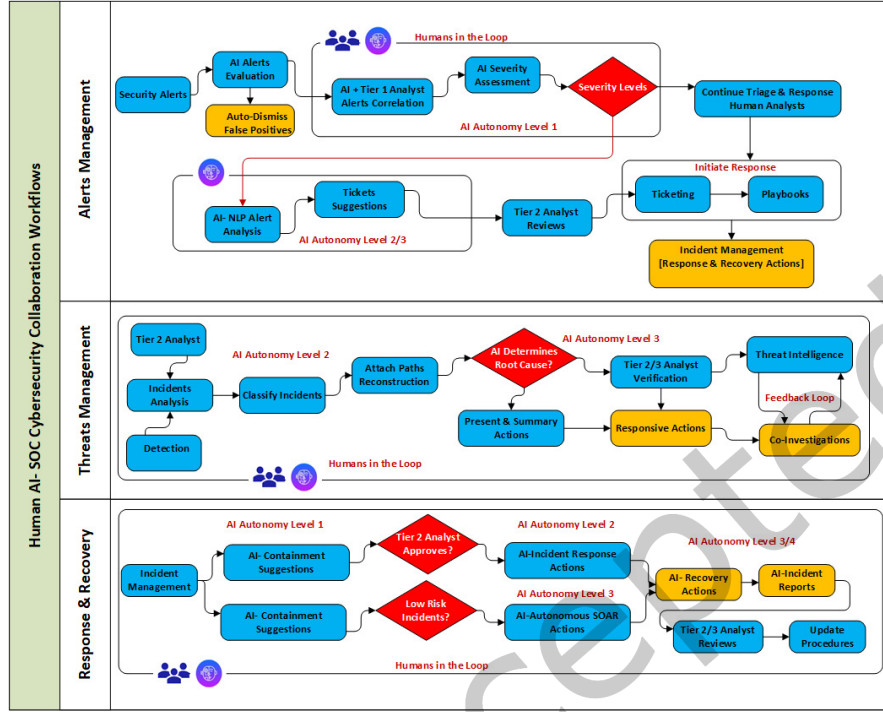


Fig. 4. Operational Workflows of Human-AI Collaboration in SOC: AI autonomy levels guide how alerts, incidents, and response actions are shared and executed between analysts and AI agents across key functional domains.

Tier 4: Governance, Oversight, and Strategic Risk Management (Govern/Recover). Tier-4 personnel assess policy and regulatory implications, coordinate incident reporting, and update organizational response frameworks [43]. At this level, AI supports tasks such as audit logging, compliance validation, and policy review [1]. Operating in HOoTL mode, analysts validate AI-generated reports and enforce standards on explainability, risk accountability, and legal compliance.

This mapping illustrates how the framework enables effective human-AI collaboration across SOC tiers. By aligning functional roles with autonomy levels and HITL configurations, it seeks to reduce Mean Time to Respond (MTTR), enhance situational awareness, and strengthen trust in AI-supported operations while maintaining transparent human oversight and accountability [1, 22, 51].

4.7 Collaborative SOC Workflows

The proposed Human-AI Collaboration Framework enables dynamic, tiered SOC workflows that adapt to varying levels of AI autonomy and trust. As AI capabilities scale, task delegation shifts from manual processes to conditional or autonomous execution, while human analysts retain oversight and strategic control. Figure 4 illustrates collaborative workflows across three core SOC functions: (i) Alerts Management, (ii) Threat Management, and (iii) Response & Recovery. System protection is omitted for simplicity, as it aligns clearly with defined human and AI roles. Each domain highlights how AI supports or automates tasks based on autonomy and complexity.

(i) **Alerts Management.** The Alerts Management workflow initiates with AI agents operating at Level 1 autonomy, assisting Tier-1 analysts by correlating alerts, performing preliminary severity assessments, and supporting

triage decisions. Humans remain actively involved in validating the AI's output and determining the appropriate response path based on alert severity. At Levels 2 and 3, AI systems advance to auto-dismiss false positives using contextual patterns, aided by NLP-based alert analysis. These systems generate ticket suggestions and recommend response actions, which are reviewed by Tier-2 analysts. At this stage, AI not only clusters and summarizes related alerts into incident candidates but also refines its prioritization through continuous feedback loops. Despite increased autonomy, human analysts remain "in-the-loop" to automatically dismiss alerts, validate critical escalations, and ensure alignment with the operational context. Human judgment is especially vital in ambiguous cases and for triage escalation. This layered collaboration improves detection fidelity, reduces analyst fatigue, and fosters trust through transparency, iterative validation, and autonomy scaling aligned with risk.

(ii) *Threat Management.* In the Threat Management workflow, AI agents operate at Level 2 autonomy to assist Tier-2 analysts in classifying incidents and reconstructing attack paths through automated log correlation and contextual analysis. This reduces investigation time and supports evidence collection for emerging threats. At Level 3, AI performs root cause analysis and generates summary reports with proposed responsive actions. Tier-2 and Tier-3 analysts remain "in-the-loop," verifying AI conclusions, incorporating operational context, and resolving ambiguities, such as differentiating legitimate administrative activity from malicious behavior. Verified intelligence is then used to drive mitigations and update detection models. This co-investigative loop, driven by human validation and AI-generated insights, enhances investigative depth, accelerates incident resolution, and supports adaptive learning. Feedback mechanisms between analysts and AI systems ensure continuous model refinement and increased precision in threat characterization and response planning.

(iii) *Response & Recovery.* The framework uses tiered autonomy to enhance containment, response, and recovery. At Level 1, AI suggests actions such as host isolation, needing Tier-2 analyst approval. Level 2 lets AI execute predefined responses via SOAR for moderate risks, post-analyst confirmation. Level 3 has AI run SOAR playbooks for low-risk events, escalating only critical cases. Analysts maintain oversight with real-time monitoring. This setup cuts response times and limits threat damage. Level 4 AI manages full recovery, including system restoration and reports, with analysts in supervisory roles. Advanced AI at Level 3 employs machine learning and anomaly detection to find hidden threats, with Tier-3 analysts validating outputs and refining AI models. This collaboration ensures quick, scalable, and reliable threat management.

Tier-4 analysts serve as strategic overseers, ensuring policy alignment, compliance, and enterprise-level risk management. They remain fully "in-the-loop," validating AI-generated reports, refining governance rules, and supervising complex recovery procedures that require organizational judgment. Their responsibilities complement the operational duties of Tiers 1–3 by ensuring that autonomy scaling remains aligned with regulatory and business constraints. Figure 4 illustrates how the Human-AI collaboration pathway, which includes permissible interaction flows, HITL, HOTL, and HOO TL, is determined by task complexity, trust, and risk, as formalized in Sections 3.2 and 4.1. The figure conceptually maps SOC tiers, autonomy levels (0–4), and oversight modes, highlighting how collaboration adapts as autonomy increases.

5 Related Work and Comparative Analysis

There is a growing body of research on introducing AI into SOC operations and on frameworks for human-automation collaboration. Here, we review and compare related works from the literature, focusing on how they address core SOC functions, levels of autonomy, and aspects of Human-AI teaming or collaboration. Table 6 summarizes key works in terms of their coverage of SOC security functions (Protection, Ingestion & Logging, Threat Detection/Management, Alerts Management, Incident Response/Recovery), the autonomy level they primarily target, and their consideration of Human-AI collaboration (including any frameworks or models proposed).

Table 6. Comparative analysis of related work on AI in SOC and Human-AI collaboration.

Legend: ✓ = partially supports; ✓✓ = moderately/full supports; ✓✓✓ = comprehensively supports; ✗ = not addressed. SOC Functions: Sys. Mon. = System Monitoring (Protection + Ingestion & Log Management); Thr. Mgt. = Threat Management (detection/hunting); Alrt. Mgt. = Alerts Management (triage/analysis); IR&R = Incident Response & Recovery. Autonomy: primary autonomy level or range targeted (0 = manual, 4 = full AI).

Autonomy	Sys. Mon.	Thr. Mgt.	Alrt. Mgt.	IR&R	Human-AI Collaboration Focus	Authors (Year)
Level 0-3	✗	✓	✓	✓	Human factors, LOA framework	Tilbury & Flowerday (2024)
Level 1-3	✗	✓	✓	✓	Autonomy, collaborative SA	Chhetri et al. (2024)
Level 1-2	✓	✓✓	✗	✗	Reciprocal learning, HITL	Cohen et al. (2025)
Level 3	✗	✓✓	✗	✓	HITL, cognitive modeling	Aref et al. (2025)
Level 2-4	✓	✓	✓	✗	AI agents for triage, and automation	Mareedu (2025)
Level 2-3	✓	✓	✗	✗	LLM-assisted threat detection & protection	Paduraru (2024)
Level 0-4	✓✓✓	✓✓✓	✓✓✓	✓✓✓	Autonomy, Trust, HITL, and SA	Our work

Prior studies highlight AI's role in enhancing analyst efficiency with human oversight in SIEM and SOAR, facilitating semi-automated decision-making [22, 24, 48]. Jalalvand *et al.* [20] studied AI-driven alert prioritization in SOC through adaptive, explainable models using contextual and historical data. Chhetri *et al.* in [10] proposed a Human-AI framework emphasizing shared awareness and trust to reduce alert fatigue and improve decision-making efficiency. Tilbury *et al.* [51] developed an automation approach for cybersecurity, introducing a SOC automation matrix for balanced Human-AI collaboration. Most SOC are minimally automated, heavily dependent on analysts, with recommendations against full automation, supporting gradual transitions to Human-in-the-Loop operations. Binbeshr *et al.* [8] highlighted AI's prevalence in SOC, as machine learning offers enhanced accuracy and fewer false positives. However, it faced challenges in data quality and model interpretability, necessitating human oversight. Tilbury and Flowerday [50, 51] perform a scoping review on *human-automation augmentation in SOC*, identifying areas of automation and issues such as automation bias and analyst complacency. Their key contribution, the SOC Automation Matrix, categorizes four human-automation relationship levels and incorporates Levels of Automation (LOA). They emphasize that while automation can cover the incident lifecycle, managing socio-technical factors (such as analyst trust) is essential, advocating for Human-in-the-Loop controls and offering guidance on aligning automation levels with tasks. Our framework complements their work by centralizing trust in Human-AI collaboration.

On the frameworks side, Chhetri *et al.* [10] present the A²C (Automate-Augment-Collaborate) framework to tackle alert fatigue. It enables dynamic switching between three modes: automated (AI handles routine alerts), augmented (AI offers suggestions to speed up analysts' decision-making), and collaborative (humans and AI jointly explore complex threats). Their vision is a flexible SOC in which the level of AI autonomy can be adjusted up or down depending on the context, which resonates with our approach. They report that implementing A²C in a prototype reduced false-negative incidents while keeping analysts in control of novel situations. Another notable work is by Cohen *et al.* [12], who focus on *Human-AI collaboration for cyber threat intelligence (CTI)*. They introduce a Reciprocal Human-Machine Learning (RHML) model, wherein two analysts and an AI system iteratively learn from each other to classify hacker forum messages. Over time, both detection accuracy and the understanding of the threat landscape by human experts improved. This demonstrates the power of continuous learning in Human-AI teams and provides a concrete example of Human-in-the-Loop machine learning in SOC-like tasks. Aref *et al.* [6] developed a Cognitive Hierarchy Theory-driven Deep Reinforcement Learning (CHT-DQN) framework for modeling Human-AI collaboration in cloud SOC. It integrates cognitive modeling, attack graphs, and HITL design for real-time threat mitigation. However, the focus remains primarily on threat

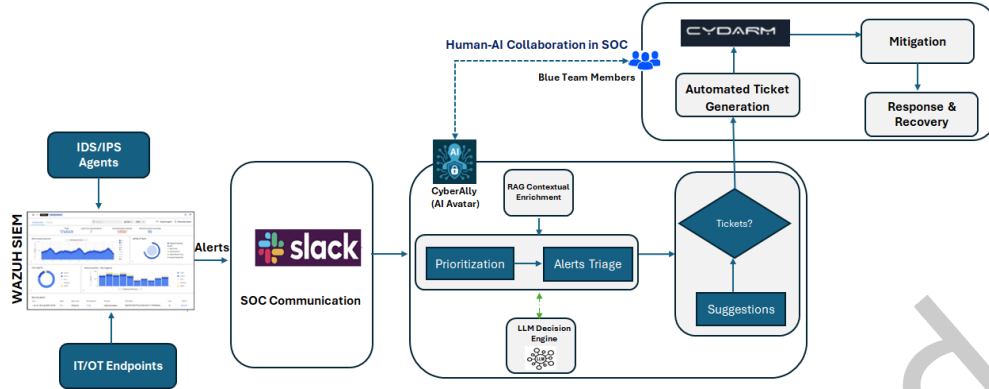


Fig. 5. Human-AI collaboration in a simulated SOC environment integrated into the ACDC Cyber Range. The CyberAlly (Avatar) AI-augmented workflow supports the Blue Team by combining real-time alert triage, LLM-based analysis, and automated ticketing.

management, with limited attention to alert management, system monitoring, and incident response. Mareedu [31] presents an autonomous-SOC vision centered on AI agents for triage and response, emphasizing orchestration through reinforcement learning, multi-agent coordination, and ethical AI governance. Kotilingala [26] focuses on LLM-enhanced threat detection pipelines within SOC workflows, improving situational awareness, precision, and proactive incident prediction. Paduraru *et al.* [44] describe a LLM-based interactive assistant for cybersecurity analysts that supports explainability, collaborative reasoning, and task-oriented decision-making.

Our framework extends existing research by integrating autonomy levels (0–4), Human-in-the-Loop (HITL) roles, and collaboration principles across key SOC functions, emphasizing trust balance and shared situational awareness. Unlike prior work focused on isolated components (e.g., CTI or vulnerability management), it provides a unified Human-AI collaboration model spanning alert management to incident response and recovery, validated through an empirical case study (Section 6).

6 Case Study: CyberAlly for SOC

We demonstrate the Human-AI Collaboration Framework through a case study in the ACDC cyber range SOC, where CyberAlly supported SIEM/EDR telemetry analysis, SOAR playbooks, and red-blue exercise control. The evaluation used ACDC datasets with ground-truth labels from structured exercises and builds on the CyberAlly AI Avatar developed under ACDC [23]. In the initial “early stages” of the simulation, analysts retained full oversight to validate AI behavior. CyberAlly’s consistent performance led analysts to grant more autonomy. These stages are trust-calibration phases in the simulation, separate from the multi-year data collection. ACDC wargame events are high-fidelity experiments with 25–40 tickets each, isolating AI-assisted workflow effects under realistic analyst loads.

6.1 Human-AI collaboration: Wargames Simulation Setup

A representative SOC environment was leveraged to mirror a mid-sized enterprise network, integrating key SOC components. Wazuh serves as the SIEM/EDR platform, aggregating security data from servers, endpoints, and network devices. We report on simulated wargames featuring CyberAlly, an AI Avatar built on a RAG augmented LLM (based on GPT models) for cybersecurity. CyberAlly integrates with SOC tools, querying SIEM, triggering SOAR workflows, and communicating via chat. Informed by simulated cyber incidents and ACDC exercises [23], it functions in wargames with 4 Red and 10 Blue Team members. A *Retrieval-Augmented Generation*

(RAG) mechanism enables queries to a security database, covering MITRE ATT&CK, threat feeds, and network topology. Knowledge graphs map entities, reducing LLM errors. Blue Team Analysts use² SOC dashboard for insights and alerts. Operating at Level 2 autonomy, CyberAlly manages events, requiring approval for major actions. Figure 5 illustrates the CyberAlly-augmented SOC workflow. Security alerts from endpoints and IDS/IPS agents are processed through Wazuh into a Slack hub for visibility. CyberAlly prioritizes and triages alerts using knowledge graphs and LLMs. Benign alerts are archived; critical ones get collaborative review, refining AI performance. For severe cases, it automates ticket creation in Cydarm, aiding mitigation. Actions and feedback are logged for enhanced learning and efficiency.

Attack Scenarios. During the ACDC cyber wargames, three attack scenarios disrupted maritime port IT and OT systems. The Red Team used the cyber kill chain and MITRE ATT&CK frameworks in the ACDC SOC, with AI support from the CyberAlly AI Avatar. Detailed Red-Team attack protocols, timing, and phase logs for Events 3 and 4 are provided in Appendix B, with full artifacts available on GitHub³. The hybrid IT-OT ACDC Port system includes 55 VMs for business, public services, IoT devices, and ICS logistics. Here are brief descriptions of the simulated scenarios:

Scenario 1: IoT Depth Sensors Attack. The first scenario involves an attack on the port's 20 km channel operations, which rely on IoT-based depth sensors for vessel coordination. These sensors collect tidal data and share it with stakeholders, the harbor master, and shipping companies via a web portal. The Red Team aims to compromise IoT depth sensors and backend systems to inject false data and disrupt transmissions, risking unsafe navigation or halting operations. The Blue Team ensures IoT data integrity and availability and maintains services with the Weather Central server.

Scenario 2: Conveyor Belt Attack. The second scenario involves compromising the port's OT loading system, specifically the conveyor belt used for loading iron ore. Controlled by OT components (PLCs, HMIs, and SCADA)⁴ When integrated with IT networks, the Red Team attempts to manipulate system logic to disrupt OT controls, causing safety hazards and loading delays. This underscores the protective challenges of mission-critical OT assets interconnected with enterprise services.

Scenario 3: Autonomous Train Supply Chain Attack. The final scenario targets the autonomous train system transporting materials to the port. Managed by an OT system controlling speed, braking, and unloading, the Red Team seeks to derail train operations, interrupt unloading, and affect supply chains. This scenario underscores cybersecurity concerns in interconnected, automated port operations.

6.2 Performance Analysis of Human-AI Collaboration in SOC Tasks

Trust-Risk-Autonomy Analysis. We begin by analyzing the (Trust-Risk-Autonomy) relationship using the formal definitions introduced in Sections 3 and 4. These equations articulate how trust (T), derived from explainability (E), performance (P), and uncertainty (U), interacts with task complexity (C) and operational risk (R) to compute the optimal autonomy level (A^*). Empirical mappings and normalization procedures are detailed in Appendix A. Figure 6 visualizes these relationships for two SOC wargame datasets (Events 3 and 4). In Event 3, the simulations show that trust scores in the upper observed range ($T \approx 0.70-0.80$) correspond to comparatively lower operational risk values ($R \approx 0.45-0.75$). This demonstrates CyberAlly's ability to support Human-on-the-Loop control, where autonomy remains at intermediate levels and analysts retain supervisory validation despite strong confidence in system outputs. Event 4 exhibits a different pattern: trust values are lower ($T = 0.66-0.75$) and operational risk is higher ($R = 0.60-0.80$), reflecting increased environmental uncertainty during this wargame. Nevertheless, autonomy remains consistently high ($A^* = 0.78-0.90$), indicating that the

²Slack for coordination during wargames: <https://slack.com>

³Human-AI Collaboration Repository

⁴, these are Maritime Port system OT assets that are part of wargame exercises.

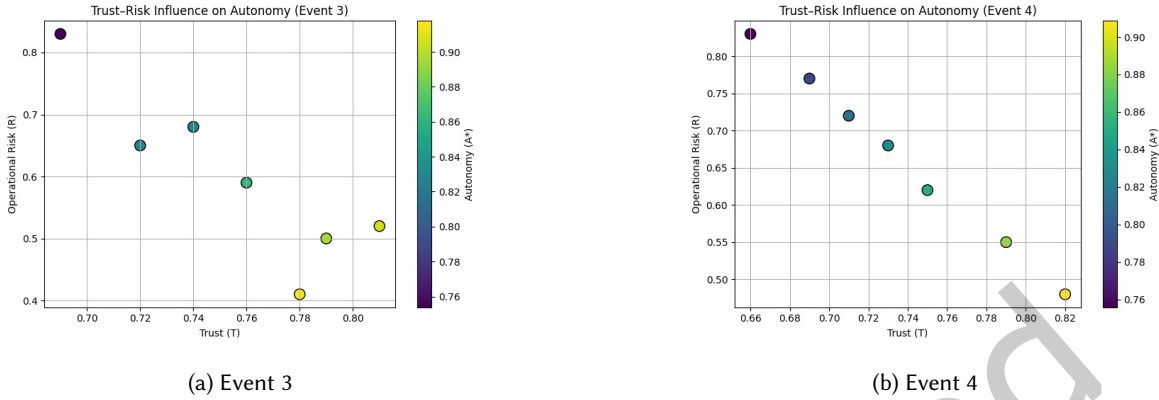


Fig. 6. Trust-Risk influence on autonomy (A^*) across two wargame events. Each point is a ticket; color encodes normalized autonomy.

framework adapts effectively to elevated risk conditions without degrading decision support. Some observed relationships arise from structural dependencies within the model (e.g., shared components influencing trust and autonomy) and should therefore be interpreted as model-consistent behavior rather than independent empirical correlations. These findings show that calibrated trust and controlled risk sustain stable autonomy levels under varying conditions. Trust in CyberAlly evolved through analyst validation, where explainability, performance consistency, and reduced uncertainty reinforced confidence and enabled effective human-AI collaboration. The results are based on balanced λ_1 and λ_2 settings; alternative configurations would shift the autonomy-risk trade-off and lead to different system behaviors. We proceed with a deeper examination of detection triage, investigation, and incident response performance below.

Detection and Triage. In the simulated ACDC Cyber Range SOC, CyberAlly operates as an LLM-enhanced agent that monitors high-volume SIEM and EDR alerts, classifying and scoring events in real time to support detection, automated ticketing, and SOAR playbook activation. At Level 1 (*Full HITL*), it functions as an alert-filtering assistant, flagging anomalies for analyst validation. During a Red Team reverse shell attack, it identified phishing emails later confirmed by analysts, reducing false positives from 70% to 35%, compared to typical manual baselines of 80–90%. Under a Human-on-the-Loop model, CyberAlly filters routine alerts while escalating high-risk events (e.g., anomalous OT administrator logins). Events 3 and 4 (Figure 7) capture key collaboration stages: AI-assisted triage escalation and joint investigation. Resolution time definitions are provided in Appendix A. Results from Wazuh-based observations (Figures 7a and 7b) show Event 3 incidents resolved within 20–40 minutes at Tier-1/2 (Level 2 autonomy). Event 4 demonstrates similar performance for more complex threats (e.g., IoT false data injection), indicating robust detection under uncertainty. Overall, AI-driven triage improved the signal-to-noise ratio (false positives reduced from 70% to 35%) and decreased ticket-resolution time by 44–56% (Appendix B), supporting effective trust-risk-autonomy calibration.

Investigation and Contextual Correlation. In cases of credential misuse, unauthorized OT access, and PLC/HMI manipulation during conveyor belt operations, CyberAlly operated semi-autonomously (*Level 2*). It retrieved logs, analyzed historical incidents, and mapped Red Team actions to mitre ATT&CK tactics. In the conveyor belt MITM scenario, the system traced the attack from an IT phishing breach to an OT compromise involving credential dumping and remote command execution, proposing actions such as validating OT workstation logs and verifying PLC configurations. While *Tier-2 analysts* retained final decision authority, CyberAlly automated cognitively intensive tasks, including log correlation and hypothesis generation, reducing investigation time from three



Fig. 7. Visualizing SOC performance with CyberAlly across response times, task types, and operational gains during wargame events.

hours to one (67% improvement), as shown in Figure 7d. Its integration of RAG and knowledge graphs enhanced contextual awareness by incorporating asset criticality, for example, identifying that compromised conveyor belt PLCs directly affect vessel loading operations. Analysts observed that CyberAlly uncovered subtle correlations, such as links between web server activity, CVEs, and potential OT compromises. Although actions required human approval, the system accelerated root cause analysis and response planning. As shown in Figure 7c, most incidents were resolved during triage (10–30 minutes) and containment (30–120 minutes), with CyberAlly further streamlining workflows by highlighting relevant MITRE ATT&CK tactics across diverse threat scenarios.

Response and Recovery Efficiency Gains. With trust, CyberAlly was allowed *autonomous* actions under *human-on-the-loop* conditions. In the train supply chain scenario, it detected abnormal commands affecting OT components for speed regulation and unloading. CyberAlly alerted engineers, flagged signals, and isolated PLCs before suggesting an inspection. In loading and web-server attacks, SOAR integration was used to rapidly isolate endpoints and block malicious IPs, reducing MTTR from 8 hours to 90 minutes. Analysts supervised and adjusted AI decisions, such as reversing unintended blocks. Figure 7d shows CyberAlly’s impact, increasing automated ticketing from 10% to 75% and cutting response times by over 60%. In complex incidents such as train unloading or conveyor manipulation, steps under HITL are followed, boosting agility and resilience.

Autonomy Calibration and Trust Evolution. From a Human-AI collaboration perspective, the integration of AI Avatar (cyberAlly) into simulated SOC operations was aligned with various HITL configurations based on incident complexity. The early stages of triage employed full HITL, ensuring that each AI-flagged alert was human-reviewed. As CyberAlly demonstrated accuracy, particularly in classifying phishing emails and triaging anomalies in OT systems, its role evolved to *on-the-loop supervision* for routine classifications. In later

response phases, especially when isolating endpoints or quarantining known threats, the system operated with partial or even full autonomy under specific rules of engagement. This calibrated autonomy approach ensured efficient task delegation without sacrificing human oversight. Trust in CyberAlly was not assumed; it was earned. Analysts initially scrutinized all AI outputs. Over time, explainable decisions, high-accuracy classifications, and consistent performance encouraged analysts to treat the AI as a reliable collaborator. By the final scenario, CyberAlly autonomously managed routine SOC workflows, allowing analysts to focus on complex decision-making. Misclassifications were rare and leveraged as learning points, refining AI outputs.

7 Discussion

7.1 Lessons Learned

This research demonstrates that Human-AI collaboration enhances SOC efficiency when trust, autonomy, and human oversight are properly balanced. AI effectively handled routine triage and enrichment, while analysts provided essential contextual judgment. The following are the key lessons learned from the study:

1. *Autonomy Calibration Across SOC Functions.* A gradual, tiered escalation of AI autonomy proved essential for balancing operational efficiency with human oversight. Early triage phases employed full HITL validation, while later response actions transitioned toward partial or conditional autonomy. Routine tasks such as phishing triage and endpoint isolation were successfully automated, while critical incident responses remained under human supervision. This autonomy layering fostered analyst trust, optimized AI deployment, and minimized the risks of automation bias or decision errors, highlighting the importance of dynamic autonomy calibration based on incident complexity and trust levels.
2. *Human-AI Communication and Autonomy.* Initially, analysts struggled with AI-generated queries and responses but adapted by refining their language and strategies. This highlights the need for Human-AI co-adaptation in SOC workflows. As AI autonomy increased, communication evolved. The system initially operated at Level 1 autonomy, requiring human verification, and progressed to Level 2 for phishing and alert triage, reaching conditional Level 3 for low-risk tasks under supervision. These shifts matched SOC task complexity, proving that autonomy scaling should be adaptive, trust-based, and task-specific, ensuring efficiency while maintaining human oversight in critical decisions.
3. *Trust but Verify.* Analysts in the early stages approached AI-suggested alerts with caution, manually verifying outputs, which aligns with Level 1 autonomy (Full HITL). Over time, as the system demonstrated consistent performance in alert triage and threat prioritization, trust increased, leading to Level 2 autonomy, where analysts intervened only when necessary. Transparent justifications were crucial for building trust. This progression showed that SOC autonomy levels should adapt to AI reliability.
4. *Avoiding Information Overload.* Rather than overwhelming analysts, AI support reduced cognitive load by filtering out benign noise. However, this positive outcome required careful calibration of alerting thresholds and relevance criteria. Without tuning, an AI assistant could exacerbate alert fatigue. Iterative adjustment cycles are necessary to optimize signal-to-noise ratios based on operational contexts.
5. *Effective Training is Essential.* Extensive domain-specific adaptation of AI systems using cyber range logs, security playbooks, and knowledge graphs was crucial. While base LLM models are fluent, they may produce inaccurate assumptions about security events without contextual grounding. This reinforces that AI assistants in SOC must be tailored to the specific organizational context; relying on general-purpose models risks unreliable performance. In this study, these principles were validated through the CyberAlly simulation environment. Organizations adopting AI SOC assistants must prioritize secure, privacy-compliant enrichment of the AI with their internal datasets.

7.2 Framework Contributions

Operationalizing Autonomy and Human Oversight. The framework formalizes task division between AI and human analysts through progressive autonomy levels, enabling dynamic role adaptation in SOC operations. Routine and repetitive tasks are delegated to AI, while humans manage complex, high-risk decisions. As incidents evolve or workloads increase, task allocation adjusts in real time based on trust and uncertainty thresholds. Early stages emphasize strict HITL supervision (Level 1), gradually scaling to conditional near-autonomous operation (Levels 2–4) as trust and confidence grow. This calibrated escalation ensures accountability, safety, and transparency while maintaining operational agility. AI agents must provide interpretable, evidence-based justifications, including data provenance, threat context, and uncertainty indicators to preserve trust at higher autonomy levels. The framework’s modular design supports heterogeneous SOC environments, integrating adaptive autonomy and cognitive reasoning while remaining compatible with existing SIEM and SOAR systems.

Empirical Validation Through Case Study. Empirical evidence of the framework’s performance was established using the ACDC CyberAlly simulation, which modeled a SOC with red and blue teams. A refined LLM-driven agent, CyberAlly, was integrated into the SOC’s security stack, interfacing with Wazuh SIEM, SOAR workflows, and a MITRE ATT&CK-grounded knowledge base. Operating at Level 2 autonomy, CyberAlly performed event monitoring, triage, and enrichment while communicating findings to analysts via natural language, with human oversight retained for critical actions. The deployment yielded substantial improvements: a **50%** reduction in false positives, **67%** faster investigation, and an **80%** decrease in Mean Time to Respond (MTTR). Analysts also reported reduced cognitive load and improved situational awareness, enabling a greater focus on strategic tasks such as threat hunting and proactive defense. Transparency and explainable outputs were identified as key factors in trust formation.

Human-AI Collaboration Workflows. These outcomes empirically substantiate the effectiveness of the proposed (Autonomy, Trust, HITL) triadic model, demonstrating measurable improvements across SOC workflows compared with traditional manual operations. The case study confirms that scalable autonomy, reinforced by explainability and human oversight, optimizes analyst task allocation, reduces fatigue, and fosters accountable Human-AI collaboration that can generalize to other cybersecurity environments.

7.3 Limitations

The framework establishes a strong foundation for integrating human expertise, artificial intelligence, and cybersecurity operations within modern SOC. By emphasizing autonomy, trust, and HITL decision-making, it outlines a scalable model for cognitive SOC of the future. However, several practical limitations remain that must be addressed for real-world deployment and sustainable Human-AI collaboration effectiveness. These limitations can be broadly categorized into four dimensions:

1. *Technical and Model Governance.* Continuous *model retraining* is required to integrate evolving threat intelligence but introduces governance challenges related to accountability, data provenance, and reliability. Managing *model drift*, where performance degrades over time, requires robust feedback loops and traceable retraining pipelines to maintain transparency and contextual relevance.
2. *Human-AI Cognitive Asymmetry.* Differences between human and AI cognition, intuition, contextual judgment, and reasoning transparency create alignment challenges. Bridging these *asymmetries* requires explainable interfaces and calibrated trust mechanisms to prevent automation bias or undertrust. Analyst tools must support rapid validation of AI-driven actions under pressure, converting outputs into interpretable insights.
3. *Organizational and Sociotechnical Readiness.* Non-technical factors such as role displacement fears, limited AI literacy, and unclear accountability hinder adoption. Gradual deployment, targeted training, and shared

decision-making are essential to build trust. The framework's success depends on sociotechnical readiness and collaboration among analysts, engineers, and governance teams for responsible AI integration.

4. *Adversarial Robustness Considerations.* As AI systems assume greater autonomy, adversarial robustness becomes critical. Threats, including prompt injection, data poisoning, and model manipulation, pose risks to SOC reliability. The framework mitigates these risks through controlled retrieval, augmented pipelines, input sanitization, output validation, differential fine-tuning with provenance-verified datasets, and periodic red-team validation of LLM modules. Future work will incorporate adversarial training datasets and continuous model-governance mechanisms to improve resilience and trustworthiness. Addressing these limitations will guide research towards adaptive autonomy, calibrated trust, and transparent Human-AI collaboration in SOC environments, informing future work directions.

8 Conclusion and Future Work

This paper presents a structured framework for Human-AI collaboration in SOCs, addressing limitations of automation-centric approaches. By integrating AI autonomy, Human-in-the-Loop (HITL), and trust across SOC tiers, the framework enables adaptive, explainable decision-making while preserving human oversight. The case study demonstrates improved operational efficiency, reduced analyst fatigue, and enhanced response coordination, thereby supporting the development of next-generation Cognitive SOCs in which AI augments human expertise. Future work will explore AI agent teams coordinated through orchestration mechanisms [11, 27], focusing on communication, redundancy reduction, and multi-agent workflows. Further research is needed to evaluate distributed AI systems through realistic simulations. LLMs offer strong potential for log analysis and incident reporting but require domain-specific fine-tuning and validation. Techniques such as RAG and datasets like cyberLLMInstruct [14] can improve reliability, while continual learning and verification remain critical [21].

Advancing Human-AI collaboration requires improved trust calibration through explainable interfaces, such as confidence scoring and interpretable reasoning [46, 58]. Future research should also investigate adaptive autonomy models, where AI control dynamically aligns with task complexity and risk, supported by reinforcement learning and decision processes [12, 18, 57]. Integrating these capabilities into governance frameworks, including the NIST AI Risk Management Framework, is essential to ensure accountability. As Agentic AI evolves, its secure integration under human oversight will be critical for building resilient and trustworthy SOC systems.

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APPENDICES

A Appendix A. Empirical Derivation of Framework Components and Dataset Mapping

Appendix A outlines how the theoretical constructs defined in Sections 3 and 4.1 are operationalized and instantiated in the CyberAlly case study (Section 6). The derivations build upon the formal foundations expressed in Equations 7, 6, and 4, which collectively describe how *task complexity* (C), *operational risk* (R), and *trust* (T) interact to dynamically calibrate AI autonomy (A) within Human–AI collaboration loops. In the CyberAlly case study, the framework variables are operationalized using normalized measurements derived from SOC operational data and analyst interaction records. Specifically, C and R are estimated from normalized operational metrics, while E and U are derived from textual and statistical measures, and P and T from analyst feedback scores, reflecting observed system performance and confidence levels. These mappings ensure that the framework’s theoretical components are quantitatively grounded, enabling reproducibility of results and providing a template for extending the analysis to other SOC environments. Each variable is summarized in Table 7, which bridges conceptual definitions with their empirical computation.

This mapping demonstrates how the theoretical constructs of the proposed Human–AI Collaboration Framework are operationalized into measurable components within SOC environments. By linking conceptual definitions (Sections 3 and 4.1) with empirical computation in the CyberAlly case study (Section 6), this appendix provides a reproducible methodological reference for trust–risk–autonomy analysis. Raw values for complexity (C) and risk (R) are first computed from observable SOC attributes (e.g., duration, severity, and operational phase). Min–max scaling is applied only as a normalization step to map these values to the range $[0, 1]$, ensuring comparability across variables. Subjective variables are derived from analyst feedback collected during cyber-range experiments using structured prompts. Specifically, perceived reliability (P) is obtained using the prompt: “How reliable was the AI recommendation for supporting your decision?”, and uncertainty (U) using: “How uncertain or inconsistent did the AI output appear?”. Responses are provided on a 1–5 scale and normalized as described in Table 7.

Explainability (E) is estimated using normalized response length as a structural proxy for explanation richness. While this does not fully capture semantic interpretability, it provides a consistent and reproducible approximation across tasks. Accordingly, E , P , and U are treated as operational proxies for transparency, perceived reliability, and consistency, enabling real-time estimation of trust (T). Trust (T) is modeled as a task-level operational index rather than a psychometric construct, reflecting the need for continuous, time-sensitive trust calibration during incident response. Finally, perceived reliability (P) influences autonomy (A) only indirectly through trust (T), while risk (R) is computed solely from incident characteristics (via I and P_{th}), ensuring separation between task-driven and perception-driven factors.

Table 7. Mapping of framework variables to conceptual definitions and empirical computation used in the CyberAlly case study.

Variable	Framework Definition	Key Features	Computation Method	Interpretation and Notes
C (Complexity)	Analytical and cognitive workload of a SOC task, determined by correlated data sources, decision depth, and task duration.	<i>created, modified, status</i>	Compute duration ($t_{mod} - t_{cre}$) and normalize to $[0,1]$ using min-max scaling.	Represents analyst workload; higher duration $\Rightarrow$ higher complexity.
R (Risk)	Expected impact of delayed or erroneous response, defined as $R = I \times P_{th}$.	<i>status, severity, org</i>	Impact I is derived from severity and operational phase (Alert=0.8, Analysis=0.6, Closed=0.2); combined with threat likelihood and normalized after raw computation .	Captures operational exposure and incident criticality.
E (Explainability)	Transparency and interpretability of AI recommendations.	<i>feedback text</i>	$E = \text{text length}/\text{max length}$ (used as a structural approximation).	Higher E indicates richer explanation; supports analyst understanding.
U (Uncertainty)	Perceived ambiguity or inconsistency in AI outputs.	<i>rate (1-5)</i>	$U = 1 - \text{std}(\text{rate})/5$. Ratings are obtained from analyst prompts assessing output certainty .	Lower U indicates more consistent outputs; $(1 - U)$ reflects confidence.
P (Perceived Reliability)	Analyst-perceived dependability of AI recommendations.	<i>rate (1-5)</i>	$P = \text{mean}(\text{rate})/5$, based on analyst assessment prompts on recommendation reliability .	Reflects experiential trust in AI-supported decisions.
T (Trust)	Aggregated analyst confidence in AI transparency, reliability, and consistency.	Derived from E, P, U	$T = 0.2E + 0.4P + 0.4(1 - U)$.	Operational trust index for SOC tasks (not a full psychometric measure).
A (Autonomy)	Degree of AI decision authority and independence from human oversight.	Derived from C, R, T	$A = 1 - (0.4C + 0.6R)(1 - T)$.	Balances task complexity, risk, and trust; higher T enables greater autonomy.

Variables Derivation Procedure. The variables C, R, E, P , and U are derived from CyberAlly SOC data using a structured, step-wise procedure aligned with Table 7. The goal is to translate raw SOC observations into normalized variables suitable for trust-risk-autonomy modeling.

Step 1: Raw Data Extraction. Structured SOC records (timestamps, alert status, severity, and organizational context) and analyst interaction data (feedback ratings and AI-generated explanations) are extracted from the cyber-range dataset.

Step 2: Task-Level Metric Computation. For each incident:

- Complexity (C): computed from investigation duration ($t_{mod} - t_{cre}$).
- Risk (R): computed as $R = I \times P_{th}$, where impact (I) is derived from severity and operational phase weighting.
- Explainability (E): computed from normalized explanation length.
- Perceived Reliability (P): computed as the mean of analyst reliability ratings.
- Uncertainty (U): computed from the dispersion (standard deviation) of analyst ratings.

Step 3: Normalization. All variables are mapped to the range $[0, 1]$ using min-max scaling. This normalization is applied only after raw values are computed to ensure comparability across heterogeneous metrics.

Step 4: Derived Variables. Trust (T) is computed from E, P , and U using Eq. 7, capturing transparency, perceived reliability, and consistency. Autonomy (A) is then computed using Eq. 4, combining task-driven factors (C, R) with perception-driven trust (T).

Step 5: Aggregation and Analysis. The resulting variables are aggregated across incidents and used to analyze Human-AI collaboration performance, as reported in Section 6.

False Discovery Rate (FDR) Computation. To measure SOC alert quality, we use the false discovery rate:

$$\text{FDR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Positives}}.$$

This reflects the analyst-facing proportion of alerts incorrectly flagged as incidents by automated systems.

SOC Timing Metrics Definition. To clarify the interpretation of SOC performance results, three timing variables are defined: investigation time, response time, and resolution time. Investigation time refers to the duration required for alert analysis, contextual correlation, and root-cause identification, reflecting analytical workload. Response time captures the time required to execute containment and mitigation actions following threat confirmation, including both manual and automated responses. Resolution time represents the total end-to-end duration required to close an incident, encompassing both investigation and response phases, along with minor operational overheads. Thus, resolution time can be expressed as an aggregate measure:

$$\text{Resolution Time} \approx \text{Investigation Time} + \text{Response Time}.$$

This decomposition enables a structured evaluation of SOC performance by distinguishing analytical efficiency (investigation), operational execution (response), and overall incident handling effectiveness (resolution), as reflected in Figure 7.

Performance Metrics. The table below reports the SOC performance metrics before and after AI integration for Events 3 and 4. The values represent aggregated operational metrics derived from the cyber dataset. Figure 7d presents normalized percentage improvements to illustrate relative performance gains across events.

Table 8. SOC Performance Metrics Before and After AI Integration (Events 3 and 4)

Metric	Event 3 (Before AI)	Event 3 (After AI)	Event 4 (Before AI)	Event 4 (After AI)
False Positive Rate (FPR)	0.70	0.35	0.72	0.38
Investigation Time (minutes)	34	9	38	12
Response Time (minutes)	40	12	45	15
Automated Ticketing Rate (%)	10%	75%	12%	70%

Dataset Description. The dataset used in this study is publicly available at the Human-AI Collaboration Dataset Repository. It is derived from cyber range-based SOC simulations under the ACDC project, incorporating Red and Blue Team wargame scenarios. The dataset includes structured and unstructured security data such as event logs, alerts, incidents, and analyst actions, supporting the evaluation of Human-AI collaboration in SOC environments. A detailed description of the dataset structure, components, and fields is provided in the Dataset Documentation.

B Appendix B. Supplementary Visualizations of SOC Performance

To complement the performance analysis presented in Section 6, Figure 8 provides a comparative visualization of SOC ticket-resolution times *before* and *after* the integration of the AI Avatar (CyberAlly). This figure demonstrates how AI-assisted triage and investigation substantially reduced mean resolution times across both events, validating the quantitative improvements discussed in Section 6 and aligning with the trust-risk-autonomy relationships analyzed in Section 4.1. Shorter resolution times reflect CyberAlly’s capability to automate detection, contextual correlation, and SOAR-triggered containment while maintaining human oversight.

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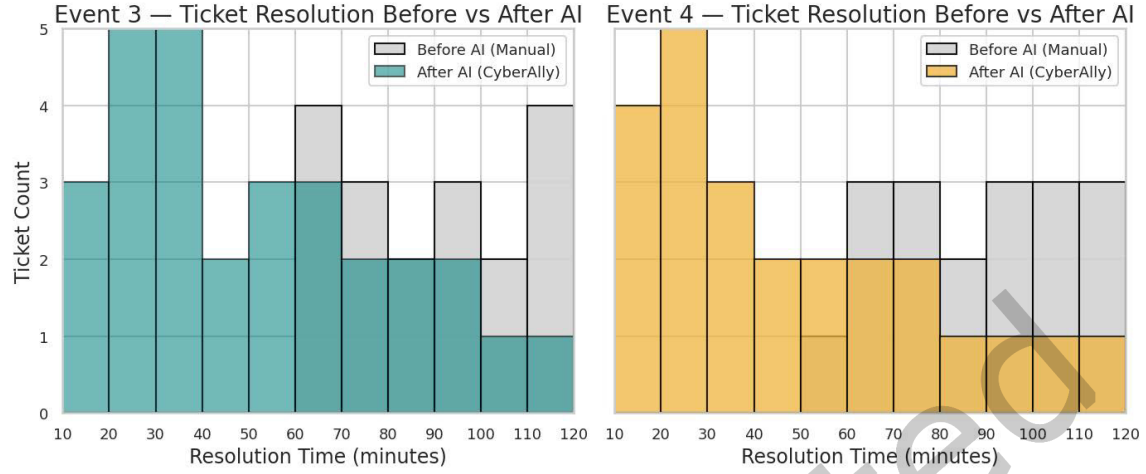


Fig. 8. Comparison of ticket-resolution times before and after AI integration for Event 3 and Event 4. Gray bars indicate manual SOC operation (Before AI), while colored bars show CyberAlly-assisted performance (After AI). Average resolution time decreased from approximately 80 minutes to 35–45 minutes, corresponding to a 44–56% improvement in incident-response efficiency.